HCIA-AI V4.0 Certification Practice Exam • Chapter 3

Module 3: Deep Learning & Foundation Models (100 Questions & Answer Key)

Curated by: Umed Jamal Nouri • v1.1 (Nov 2026) • github.com/UmedJaMala

Question 1

TRUE OR FALSE • 16 Points

Typically, in a deep neural network that uses the Sigmoid activation function, the gradient may decrease close to 0 within five layers, causing severe vanishing gradients.

A. True
B. False

Question 2

TRUE OR FALSE • 16 Points

When training a deep neural network with the Softsign function, gradients never saturate regardless of how deep the network is.

A. True
B. False

Question 3

TRUE OR FALSE • 16 Points

The Tanh activation function can effectively eliminate and completely solve the vanishing gradient problem in arbitrarily deep neural networks.

A. True
B. False

Question 4

TRUE OR FALSE • 16 Points

Batch Normalization is a technique used in deep learning to standardize layer inputs, stabilizing internal covariate shift and accelerating model convergence.

A. True
B. False

Question 5

TRUE OR FALSE • 16 Points

A single-layer perceptron without hidden layers is capable of solving linearly non-separable problems such as the XOR logical operation.

A. True
B. False

Page Answers: **Q1**: True • **Q2**: False • **Q3**: False • **Q4**: True • **Q5**: False

Question 6

TRUE OR FALSE • 16 Points

The maximum derivative of the standard Sigmoid function is 1.0, which occurs when its input x equals 0.

A. True

B. False

Question 7

TRUE OR FALSE • 16 Points

The ReLU activation function outputs 0 for all negative inputs ($x < 0$) and has a derivative of 1 for all positive inputs ($x > 0$).

A. True

B. False

Question 8

TRUE OR FALSE • 16 Points

During neural network inference (testing), Dropout continues to randomly set 50% of the neurons' outputs to zero.

A. True

B. False

Question 9

TRUE OR FALSE • 16 Points

In Convolutional Neural Networks, the pooling layer introduces learnable weight parameters that are updated via backpropagation.

A. True

B. False

Question 10

TRUE OR FALSE • 16 Points

Parameter sharing in CNNs means that every single pixel in an image requires a completely unique, independent set of convolution kernel weights.

A. True

B. False

Page Answers: **Q6**: False • **Q7**: True • **Q8**: False • **Q9**: False • **Q10**: False

Question 11

TRUE OR FALSE • 16 Points

Compared with standard RNNs, CNNs are better suited for extracting 2D spatial grid features from images.

A. True
B. False

Question 12

TRUE OR FALSE • 16 Points

Transformer models discard recurrent loops completely and rely on Self-Attention mechanisms and Positional Encoding to capture token dependencies.

A. True
B. False

Question 13

TRUE OR FALSE • 16 Points

The Adam optimizer combines the principles of Momentum (first moment estimation) and RMSProp (second moment estimation).

A. True
B. False

Question 14

TRUE OR FALSE • 16 Points

LSTM networks contain only a single gate known as the Reset Gate.

A. True
B. False

Question 15

TRUE OR FALSE • 16 Points

The Gated Recurrent Unit (GRU) contains five separate gates, which makes it much more computationally complex than standard LSTM.

A. True
B. False

Page Answers: **Q11**: True • **Q12**: True • **Q13**: True • **Q14**: False • **Q15**: False

Question 16

TRUE OR FALSE • 16 Points

In ResNet, residual shortcut (skip) connections allow gradients to flow directly through identity paths, enabling effective training of networks with hundreds of layers.

A. True
B. False

Question 17

TRUE OR FALSE • 16 Points

Exploding gradients during deep network training can lead to numerical overflow and NaN loss values, which can be mitigated by Gradient Clipping.

A. True
B. False

Question 18

TRUE OR FALSE • 16 Points

The AdaGrad optimizer monotonically increases the learning rate during training, eventually causing infinite learning rates.

A. True
B. False

Question 19

TRUE OR FALSE • 16 Points

In the Transformer architecture, the Scaled Dot-Product Attention divides the dot products of Q and K by the square root of the key dimension (sqrt(d_k)) to prevent softmax gradients from becoming extremely small.

A. True
B. False

Question 20

TRUE OR FALSE • 16 Points

BERT is an autoregressive decoder-only language model trained primarily to predict the very next token in a causal sequence.

A. True
B. False

Page Answers: **Q16**: True • **Q17**: True • **Q18**: False • **Q19**: True • **Q20**: False

Question 21

TRUE OR FALSE • 16 Points

GPT models adopt a bidirectional encoder architecture where each token attends at the same time to all past and future tokens during causal generation.

A. True
B. False

Question 22

TRUE OR FALSE • 16 Points

Data augmentation techniques (e.g., random flipping, cropping, and color jittering) reduce training set size and increase model underfitting.

A. True
B. False

Question 23

TRUE OR FALSE • 16 Points

Cross-Entropy loss is widely used with the Softmax activation function for multi-class classification neural networks.

A. True
B. False

Question 24

TRUE OR FALSE • 16 Points

Backpropagation calculates the gradient of the loss function with respect to each network parameter using the mathematical Chain Rule of calculus.

A. True
B. False

Question 25

TRUE OR FALSE • 16 Points

In standard RNNs, Backpropagation Through Time (BPTT) unfolds the recurrent computational graph across time steps to compute weight gradients.

A. True
B. False

Page Answers: **Q21**: False • **Q22**: False • **Q23**: True • **Q24**: True • **Q25**: True

Question 26

SINGLE-ANSWER QUESTION • 16 Points

During neural network training, which of the following values are continuously updated by optimization algorithms using gradient descent?

- **A.** Sample labels
- **B.** Learning rate
- **C.** Model parameters
- **D.** Batch size

Question 27

SINGLE-ANSWER QUESTION • 16 Points

The ReLU function is commonly used in deep learning. What is the value range (output range) of the standard ReLU function?

- **A.** (-inf, +inf)
- **B.** (-1.0, +1.0)
- **C.** [0, +inf)
- **D.** (0.0, 1.0)

Question 28

SINGLE-ANSWER QUESTION • 16 Points

The Sigmoid activation function is monotonic and continuous, but when used in very deep networks, which problem is most likely to emerge?

- **A.** Vanishing gradient
- **B.** Gradient explosion
- **C.** Underfitting caused by high variance
- **D.** Loss function becoming negative

Question 29

SINGLE-ANSWER QUESTION • 16 Points

Which of the following activation functions can effectively reduce the vanishing gradient problem in positive activation regions?

- **A.** ReLU
- **B.** Tanh
- **C.** Softsign
- **D.** Sigmoid

Question 30

SINGLE-ANSWER QUESTION • 16 Points

Which of the following is NOT a recurrent neural network model?

- **A.** Long Short-Term Memory
- **B.** Logistic Regression Classifier
- **C.** Gated Recurrent Unit
- **D.** Bidirectional RNN

Page Answers: **Q26**: C • **Q27**: C • **Q28**: A • **Q29**: A • **Q30**: B

Question 31

SINGLE-ANSWER QUESTION • 16 Points

If a 32 x 32 feature map is processed through a 2 x 2 pooling layer with stride = 2 and no padding, what is the size of the output feature map?

- **A.** 30 x 30
- **B.** 32 x 32
- **C.** 64 x 64
- **D.** 16 x 16

Question 32

SINGLE-ANSWER QUESTION • 16 Points

If a 32 x 32 image is input and a 5 x 5 convolution kernel is applied with stride = 1 and padding = 0, what is the spatial dimension of the output feature map?

- **A.** 27 x 27
- **B.** 30 x 30
- **C.** 28 x 28
- **D.** 32 x 32

Question 33

SINGLE-ANSWER QUESTION • 16 Points

Which layer in a Convolutional Neural Network achieves translational invariance and reduces spatial dimensions without introducing trainable parameters?

- **A.** Batch Normalization Layer
- **B.** Pooling Layer
- **C.** Convolutional Layer
- **D.** Fully-Connected Layer

Question 34

SINGLE-ANSWER QUESTION • 16 Points

Which optimizer uses both the exponentially decaying average of past gradients (first moment) and the past squared gradients (second moment)?

- **A.** Standard SGD
- **B.** Adam
- **C.** Momentum SGD
- **D.** AdaGrad

Question 35

SINGLE-ANSWER QUESTION • 16 Points

In the LSTM cell, which gate is responsible for deciding what percentage of information from the previous cell state (C_{t-1}) should be discarded?

- **A.** Forget Gate
- **B.** Input Gate
- **C.** Reset Gate
- **D.** Output Gate

Page Answers: **Q31**: D • **Q32**: C • **Q33**: B • **Q34**: B • **Q35**: A

Question 36

SINGLE-ANSWER QUESTION • 16 Points

Which gate in the LSTM cell decides which newly computed candidate values should be stored in the updated cell state?

- **A.** Update Gate
- **B.** Forget Gate
- **C.** Output Gate
- **D.** Input Gate

Question 37

SINGLE-ANSWER QUESTION • 16 Points

What is the primary motivation behind the Residual Connection (shortcut connection) introduced in ResNet?

- **A.** Enabling identity mappings F(x) + x so gradients flow smoothly, preventing network degradation
- **B.** Replacing non-linear activation functions with direct linear identity pass-through layers
- **C.** Flattening two-dimensional spatial feature maps into single-dimension sequential vectors
- **D.** Doubling kernel weight parameter counts per layer to maximize model memorization capacity

Question 38

SINGLE-ANSWER QUESTION • 16 Points

In the Transformer Self-Attention formula Attention(Q, K, V) = softmax(Q K^T / sqrt(d_k)) V, what does 'd_k' represent?

- **A.** Total number of training samples
- **B.** Total vocabulary size
- **C.** Dimension of the key vector
- **D.** Number of layers in the encoder

Question 39

SINGLE-ANSWER QUESTION • 16 Points

Why is Positional Encoding essential in the Transformer architecture?

- **A.** To replace dense feedforward projection layers with fixed sinusoidal trigonometric operations
- **B.** To calculate cross-entropy loss gradients across multi-token target vocabulary dictionaries
- **C.** To provide the permutation-invariant self-attention mechanism with sequence order awareness
- **D.** To project sparse token index embeddings into low-dimensional normalized spatial coordinates

Question 40

SINGLE-ANSWER QUESTION • 16 Points

Which activation function is defined as f(x) = max(0.01x, x) to solve the 'Dying ReLU' problem?

- **A.** Exponential Linear Unit
- **B.** Scaled Exponential Unit
- **C.** Gaussian Error Linear Unit
- **D.** Leaky Rectified Linear Unit

Page Answers: **Q36**: D • **Q37**: A • **Q38**: C • **Q39**: C • **Q40**: D

Question 41

SINGLE-ANSWER QUESTION • 16 Points

What happens when a hidden neuron experiences the 'Dying ReLU' problem?

- **A.** Its weight values explode toward positive infinity, triggering numerical NaN overflow errors
- **B.** Its activation function spontaneously transitions into a continuous hyperbolic tangent curve
- **C.** Its input stays negative, causing 0 output and 0 gradient, permanently halting parameter updates
- **D.** Its internal bias coefficient inverts polarity, forcing the activation to saturate at positive one

Question 42

SINGLE-ANSWER QUESTION • 16 Points

Which loss function is mathematically defined as L = - sum(y_i * log(p_i)) for multi-class classification?

- **A.** Mean Squared Error
- **B.** Cross-Entropy Loss
- **C.** Hinge Loss
- **D.** Mean Absolute Error

Question 43

SINGLE-ANSWER QUESTION • 16 Points

In Convolutional Neural Networks, what is a 1x1 convolution commonly used for?

- **A.** Doubling the spatial height and width dimensions of feature maps through bilinear upsampling
- **B.** Cross-channel pooling to reduce or expand channel depth while injecting non-linear computation
- **C.** Extracting directional edge gradients across spatial diagonal pixel coordinates in input images
- **D.** Binarizing continuous feature activations into discrete zero-or-one boolean threshold outputs

Question 44

SINGLE-ANSWER QUESTION • 16 Points

What is the primary role of the Output Gate in an LSTM unit?

- **A.** Discarding obsolete historical context from the previous cell state vector during reset steps
- **B.** Resetting the internal forget gate bias to zero at the start of every sequence
- **C.** Controlling which filtered portion of the updated cell state is exposed as hidden state h_t
- **D.** Scaling the input embedding matrix by the inverse square root of hidden dimension

Question 45

SINGLE-ANSWER QUESTION • 16 Points

What is the key advantage of Multi-Head Attention over single-head attention in Transformers?

- **A.** It processes multiple independent batch sequences at the same time using a single shared linear matrix
- **B.** It projects inputs into diverse representation subspaces, allowing joint attention to different features
- **C.** It eliminates the requirement for positional encodings by computing temporal ordering dynamically
- **D.** It replaces floating-point matrix multiplications with low-precision integer bitwise operations

Page Answers: **Q41**: C • **Q42**: B • **Q43**: B • **Q44**: C • **Q45**: B

Question 46

SINGLE-ANSWER QUESTION • 16 Points

In deep learning, what does 'Weight Decay' directly correspond to in terms of loss regularization?

- **A.** L1 Regularization
- **B.** Dropout probability
- **C.** L2 Regularization
- **D.** Learning rate scheduling

Question 47

SINGLE-ANSWER QUESTION • 16 Points

What regularization method temporarily removes units (both hidden and visible) along with their incoming and outgoing edges with probability p during each training iteration?

- **A.** Dropout
- **B.** Gradient clipping
- **C.** Batch normalization
- **D.** Feature scaling

Question 48

SINGLE-ANSWER QUESTION • 16 Points

Which of the following describes the training procedure of the BERT foundation model?

- **A.** K-Means clustering of raw pixels
- **B.** Masked Language Modeling and Next Sentence Prediction
- **C.** Supervised reinforcement learning from human feedback
- **D.** Next token prediction only using causal masking

Question 49

SINGLE-ANSWER QUESTION • 16 Points

Which classical CNN architecture won the ImageNet Large Scale Visual Recognition Challenge (ILSVRC) in 2012, sparking the modern deep learning boom?

- **A.** ResNet-152
- **B.** Transformer
- **C.** LeNet-5
- **D.** AlexNet

Question 50

SINGLE-ANSWER QUESTION • 16 Points

What is the primary function of the 'C' (Cell State) vector in an LSTM network?

- **A.** Acting as an unobstructed conveyor belt that carries linear memory across steps with additive updates
- **B.** Calculating cross-entropy objective loss against ground truth label targets on every recurrent step
- **C.** Applying softmax normalization directly to the unscaled input gate logits at each step
- **D.** Multiplying hidden state vectors by a fixed independent rotation matrix at each step

Page Answers: **Q46**: C • **Q47**: A • **Q48**: B • **Q49**: D • **Q50**: A

Question 51

SINGLE-ANSWER QUESTION • 16 Points

In a deep neural network using the standard Sigmoid activation function sigma(x), what is the maximum value of its first derivative sigma'(x), and where does it occur?

- **A.** Maximum derivative is 0.25 (1/4), occurring at x = 0
- **B.** Maximum derivative is 1.0, occurring at x = 0
- **C.** Maximum derivative is 0.5, occurring as x approaches +infinity
- **D.** Maximum derivative is 2.0, occurring at x = 1

Question 52

SINGLE-ANSWER QUESTION • 16 Points

Why are non-linear activation functions mandatory between successive layers in deep neural networks?

- **A.** To initialize all neural weight parameters uniformly to constant values of positive one point zero
- **B.** To convert floating-point tensor activations into discrete binary booleans for hardware caching
- **C.** To prevent multiple stacked linear layers from mathematically collapsing into a single linear map
- **D.** To restrict operating system memory consumption and prevent CUDA kernel deadlocks during training

Question 53

SINGLE-ANSWER QUESTION • 16 Points

Which optimizer maintains an exponential moving average of past gradients to help accelerate in directions with persistent gradients and dampen oscillations?

- **A.** Nearest Neighbor Search
- **B.** Standard SGD without momentum
- **C.** Exhaustive Grid Search
- **D.** Momentum Optimizer

Question 54

SINGLE-ANSWER QUESTION • 16 Points

In an image with 3 color channels (RGB) of size 224 x 224, what are the dimensions of a single convolution kernel of spatial size 3 x 3?

- **A.** 1 x 1 x 3
- **B.** 3 x 3 x 3
- **C.** 224 x 224 x 3
- **D.** 3 x 3 x 1

Question 55

SINGLE-ANSWER QUESTION • 16 Points

What is the formula for calculating output spatial dimension O given input size W, kernel size K, padding P, and stride S in a CNN layer?

- **A.** O = floor((W - K + 2P) / S) - 1
- **B.** O = floor((W - K + 2P) / S)
- **C.** O = floor((W - K + 2P) / S) + 2
- **D.** O = floor((W - K + 2P) / S) + 1

Page Answers: **Q51**: A • **Q52**: C • **Q53**: D • **Q54**: B • **Q55**: D

Question 56

SINGLE-ANSWER QUESTION • 16 Points

Which of the following is a key architectural characteristic of the Swish activation function f(x) = x * sigmoid(beta * x)?

- **A.** It restricts all activated output values strictly within the closed real interval of zero and one
- **B.** It eliminates backward gradient propagation by computing forward parameter updates analytically
- **C.** It truncates all non-zero floating-point numbers into eight-bit signed integer representations
- **D.** It is smooth and non-monotonic, consistently reducing gradient vanishing in deep architectures

Question 57

SINGLE-ANSWER QUESTION • 16 Points

Which layer normalization technique in deep learning normalizes activations across features for each individual sample, widely used in Transformer models?

- **A.** Batch Normalization
- **B.** Min-Max Normalization
- **C.** Layer Normalization
- **D.** Weight Normalization

Question 58

SINGLE-ANSWER QUESTION • 16 Points

What is 'Early Stopping' in neural network training?

- **A.** Halting model training when validation performance ceases improving, thereby preventing overfitting
- **B.** Deleting intermediate training checkpoints automatically whenever the loss curve plateaus on disk
- **C.** Terminating model optimization immediately after epoch one regardless of error convergence rates
- **D.** Continuing gradient updates until training loss reaches zero across all epochs

Question 59

SINGLE-ANSWER QUESTION • 16 Points

Which architecture is specifically designed for sequence-to-sequence tasks (e.g., machine translation) using two separate RNN or Transformer blocks?

- **A.** Single Decision Tree
- **B.** Encoder-Decoder Architecture
- **C.** Linear Hyperplane Sorter
- **D.** K-Means Clusterer

Question 60

MULTIPLE-ANSWER QUESTION • 18 Points

In the Transformer architecture, which mechanisms are integrated into each sub-layer to stabilize gradient flow and accelerate deep network training? (Multiple Choice)

- **A.** Residual connection
- **B.** Max pooling
- **C.** Layer normalization
- **D.** Convolution kernel

Page Answers: **Q56**: D • **Q57**: C • **Q58**: A • **Q59**: B • **Q60**: AC

Question 61

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following factors contribute to the Vanishing or Exploding Gradient problem during backpropagation through time (BPTT) in standard RNNs?

- **A.** Repeated matrix multiplications of weight matrices across extended sequential time steps
- **B.** Use of saturating activation functions like Sigmoid whose derivatives are bounded
- **C.** Operating on fixed-width non-overlapping spatial convolutional sliding windows
- **D.** Long-range temporal dependencies that exponentially attenuate or escalate backpropagated error

Question 62

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following represent recognized methods of Parameter-Efficient Fine-Tuning (PEFT) for large foundation models?

- **A.** Full Model Parameter Retraining on entire multi-terabyte web crawl datasets
- **B.** LoRA injecting low-rank decomposition matrices
- **C.** Prefix Tuning and Prompt Tuning conditioning model activations with learnable vectors
- **D.** Re-initializing all transformer feedforward projection layers from random uniform distributions

Question 63

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements regarding LoRA (Low-Rank Adaptation) fine-tuning are correct?

- **A.** It freezes the original pre-trained model weight matrices during fine-tuning
- **B.** It completely removes the multi-head attention mechanism from Transformer layers
- **C.** It introduces trainable low-rank decomposition matrices alongside attention projections
- **D.** It greatly reduces the number of trainable parameters and GPU memory consumption

Question 64

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following activation functions have strictly bounded output ranges that squash inputs into finite numerical intervals?

- **A.** Sigmoid function
- **B.** Tanh function
- **C.** Standard ReLU function
- **D.** Leaky ReLU function

Question 65

MULTIPLE-ANSWER QUESTION • 18 Points

According to the HCIA-AI V4.0 curriculum, which of the following statements accurately describe the structure and properties of a Feedforward Neural Network (FNN)?

- **A.** Signals are transmitted unidirectionally from the input layer to the output layer with no feedback loops
- **B.** The complete network architecture can be represented mathematically as a Directed Acyclic Graph (DAG)
- **C.** The bias term 'b' acts as the threshold required for activating a neuron
- **D.** Feedback edges route output activations cyclically back to input nodes during forward inference

Page Answers: **Q61**: ABD • **Q62**: BC • **Q63**: ACD • **Q64**: AB • **Q65**: ABC

Question 66

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following activation functions are smooth, bounded S-shaped (sigmoidal) curves commonly used in neural networks?

- **A.** Sigmoid function
- **B.** Hyperbolic Tangent or Tanh function
- **C.** Standard Rectified Linear Unit
- **D.** Leaky ReLU function

Question 67

MULTIPLE-ANSWER QUESTION • 18 Points

In the Transformer scaled dot-product attention formula, which vectors are multiplied together via dot-product to compute raw compatibility scores?

- **A.** Query vector
- **B.** Value vector
- **C.** Key vector
- **D.** Residual shortcut vector

Question 68

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following algorithms are commonly used first-order gradient-based optimizers in deep learning?

- **A.** RMSProp
- **B.** Momentum SGD
- **C.** K-Means
- **D.** Adam

Question 69

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following components are foundational architectural building blocks of Convolutional Neural Networks (CNNs)?

- **A.** Recurrent hidden-to-hidden sequential transition matrices
- **B.** Convolutional layers with learnable spatial filter kernels
- **C.** Pooling layers
- **D.** Non-linear activation functions

Question 70

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are recognized regularization techniques used to prevent overfitting in deep neural networks?

- **A.** Early stopping based on validation loss monitoring
- **B.** L1 and L2 weight decay regularization
- **C.** Data augmentation expanding training sample diversity
- **D.** Dropout randomly deactivating neurons during training

Page Answers: **Q66**: AB • **Q67**: AC • **Q68**: ABD • **Q69**: BCD • **Q70**: ABCD

Question 71

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements regarding the Adam optimizer are correct?

- **A.** It tracks an exponentially decaying average of past gradients
- **B.** It completely eliminates the necessity of error backpropagation gradient evaluation
- **C.** It tracks an exponentially decaying average of past squared gradients
- **D.** It includes bias-correction terms to counteract zero-bias initialization

Question 72

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are the primary pooling operations commonly used in Convolutional Neural Networks?

- **A.** Max Pooling
- **B.** Median Value Sorting Filter
- **C.** Average Pooling
- **D.** Cross-Entropy Loss Normalization

Question 73

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements about Long Short-Term Memory (LSTM) are true?

- **A.** LSTM introduces dedicated gate mechanisms to reduce vanishing gradients in standard RNNs
- **B.** The Cell State acts as a linear conveyor belt across long time sequences
- **C.** The Forget Gate controls information retention from previous cell state
- **D.** LSTM can only be used for image classification and cannot process text

Question 74

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are key architectural advantages of the Transformer over classical RNN models?

- **A.** Ability to scale efficiently to billions of parameters
- **B.** Direct global dependency modeling without distance-related attenuation
- **C.** Eliminating the need for any training data
- **D.** Full parallelization during training across all sequence tokens

Question 75

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are classic deep Convolutional Neural Network architectures?

- **A.** VGGNet
- **B.** LeNet-5
- **C.** AlexNet
- **D.** ResNet

Page Answers: **Q71**: ACD • **Q72**: AC • **Q73**: ABC • **Q74**: ABD • **Q75**: ABCD

Question 76

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following structural mechanisms are integrated inside a standard Transformer Encoder block?

- **A.** Multi-Head Self-Attention layer
- **B.** Depthwise spatial 2D max-pooling subsampling
- **C.** Layer Normalization and residual shortcut connections
- **D.** Position-wise Feedforward Neural Network

Question 77

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are recognized mathematical limitations of the classic Sigmoid activation function?

- **A.** Vanishing gradient problem when inputs have large absolute values
- **B.** Outputs are not zero-centered, causing zigzag gradient updates during backpropagation
- **C.** Output values unbounded toward positive infinity
- **D.** Complete absence of first-order mathematical differentiability

Question 78

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are typical foundation model methods?

- **A.** Autoregressive language models
- **B.** Single-perceptron linear classifiers
- **C.** Encoder-decoder sequence models
- **D.** Masked autoencoder language models

Question 79

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements about Dropout are correct?

- **A.** During evaluation/testing, all neurons are kept active and outputs are scaled appropriately
- **B.** It randomly sets activations of selected neurons to zero with probability p during training
- **C.** It prevents complex co-adaptation of feature detectors
- **D.** It is an unsupervised clustering algorithm

Question 80

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are standard loss functions used for multi-class classification and continuous regression respectively?

- **A.** Triplet Contrastive Distance Metric
- **B.** Silhouette Cluster Separation Index
- **C.** Categorical Cross-Entropy Loss
- **D.** Mean Squared Error

Page Answers: **Q76**: ACD • **Q77**: AB • **Q78**: ACD • **Q79**: ABC • **Q80**: CD

Question 81

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements about Batch Normalization are true?

- **A.** It introduces learnable scale and shift parameters
- **B.** It normalizes each feature dimension across the mini-batch during training
- **C.** It allows higher learning rates and accelerates training convergence
- **D.** It can only be used on the final output classification layer

Question 82

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements about Backpropagation (BP) are correct?

- **A.** It generates random training labels for supervised datasets
- **B.** It computes parameter gradients backward from the output layer to the input layer
- **C.** It relies on the mathematical chain rule of differentiation
- **D.** Gradients computed by BP are passed to optimizers to update model weights

Question 83

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements correctly describe the standard ReLU activation function f(x) = max(0, x)?

- **A.** Its output is strictly bounded within the closed real range of negative one and positive one
- **B.** It possesses a smooth, non-linear curve with continuous second-order derivatives everywhere
- **C.** For positive inputs, its derivative is constant 1, preventing gradient vanishing
- **D.** For negative inputs, its output is constant 0, inducing sparse network representation

Question 84

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are classic Parameter-Efficient Fine-Tuning (PEFT) methods for adapting large pre-trained models?

- **A.** LoRA
- **B.** Prompt Tuning
- **C.** Full retraining
- **D.** Random initialization

Question 85

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following problems can occur when training very deep neural networks without appropriate normalization or skip connections?

- **A.** Faster convergence
- **B.** Vanishing gradients
- **C.** Network degradation
- **D.** Exploding gradients

Page Answers: **Q81**: ABC • **Q82**: BCD • **Q83**: CD • **Q84**: AB • **Q85**: BCD

Question 86

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements about the Softmax function are correct?

- **A.** Each output probability is strictly bounded between 0 and 1
- **B.** It converts real-valued logits into a probability distribution
- **C.** It can only be used when there are exactly two classes
- **D.** The sum of all output probabilities across classes equals exactly 1

Question 87

MULTIPLE-ANSWER QUESTION • 18 Points

Unlike LSTM which contains three gates, the Gated Recurrent Unit (GRU) simplifies recurrent gating into which distinct gates?

- **A.** Forget Gate
- **B.** Reset Gate
- **C.** Output Gate
- **D.** Update Gate

Question 88

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are benefits of Data Augmentation in Computer Vision?

- **A.** Helps prevent model overfitting
- **B.** Deletes corrupted image files from cloud storage
- **C.** Increases the diversity of training samples without gathering new data
- **D.** Improves the generalization ability of neural networks

Question 89

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are common types of Positional Encodings used to provide sequence order awareness in Transformers?

- **A.** Sinusoidal encodings
- **B.** Learned Positional Embeddings
- **C.** Stochastic Gaussian Noise Additive Perturbations
- **D.** Batch Normalization Running Variance Scaling

Question 90

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are hyperparameters of a Convolutional Layer?

- **A.** Padding size
- **B.** Kernel size
- **C.** Stride
- **D.** Number of output channels

Page Answers: **Q86**: ABD • **Q87**: BD • **Q88**: ACD • **Q89**: AB • **Q90**: ABCD

Question 91

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are primary structural layers commonly found in Convolutional Neural Networks (CNNs)?

- **A.** Convolutional Layers
- **B.** Pooling Layers
- **C.** Fully Connected Layers
- **D.** Recurrent Feedback Memory Units

Question 92

MULTIPLE-ANSWER QUESTION • 18 Points

In deep neural network training, which of the following official techniques are used to mitigate Exploding Gradients and Vanishing Gradients?

- **A.** Applying Gradient Clipping to cap gradients exceeding a threshold, preventing exploding gradients
- **B.** Applying L1 and L2 weight regularization to constrain weight growth and resolve exploding gradients
- **C.** Using ReLU activations (derivative 1 for x > 0) and LSTM gating to alleviate vanishing gradients
- **D.** Initializing every weight tensor across all hidden layers strictly to constant zero

Question 93

FILL-IN-THE-BLANK QUESTION • 16 Points

A Feedforward Neural Network (FNN) transmits signals unidirectionally with no feedback, so its topology can be represented as a Directed Acyclic Graph (abbreviation: ____).

Answer: _____

Question 94

FILL-IN-THE-BLANK QUESTION • 16 Points

In a Feedforward Neural Network neuron, the bias parameter 'b' represents the activation ____ required to fire the neuron.

Answer: _____

Question 95

FILL-IN-THE-BLANK QUESTION • 16 Points

For the standard Sigmoid activation function, the maximum value of its derivative sigma'(x) (occurring at x = 0) is ____. (Enter a decimal number.)

Answer: _____

Page Answers: **Q91**: ABC • **Q92**: ABC • **Q93**: DAG • **Q94**: threshold • **Q95**: 0.25

Question 96

FILL-IN-THE-BLANK QUESTION • 16 Points

Alongside Gradient Clipping, applying L1 or L2 weight ____ is an effective deep learning technique to mitigate the Exploding Gradient problem.

Answer: _____

Question 97

FILL-IN-THE-BLANK QUESTION • 16 Points

Given a 32 x 32 input image, a 5 x 5 convolution kernel with stride 1 and padding 0, the spatial width (and height) of the output feature map is ____. (Enter a number.)

Answer: _____

Question 98

FILL-IN-THE-BLANK QUESTION • 16 Points

On a CNN layer with 128 filters of spatial size 3 x 3 processing an input feature map of size 28 x 28 x 64, the depth of each convolution kernel is ____. (Enter a number.)

Answer: _____

Question 99

FILL-IN-THE-BLANK QUESTION • 16 Points

In Transformer Scaled Dot-Product Attention Attention(Q, K, V) = softmax((Q K^T) / sqrt(d_k)) V, the scaling factor sqrt(d_k) uses d_k, which represents the dimension of the ____ vector.

Answer: _____

Question 100

FILL-IN-THE-BLANK QUESTION • 16 Points

In multi-class neural network classifiers, the sum of all output probabilities produced by the Softmax layer across every class is always ____. (Enter a number.)

Answer: _____

Page Answers: **Q96**: regularization • **Q97**: 28 • **Q98**: 64 • **Q99**: Key (or key) • **Q100**: 1